Rheological Model Fitting

Methylated Alginate (MeAlg) Solutions — Viscometry Data (3.4.26)

Power Law Model | Carreau Model

1. Overview

Viscometry experiments were performed on MeAlg solutions at five concentrations: 6%, 8%, 10%, 12%, and 14%. The flow curve data (shear stress and viscosity vs. shear rate) were fitted to two rheological models: the Power Law model and the Carreau model. Both models were applied independently to all five concentrations.

The primary objective was to obtain the consistency index K and flow behavior index n from the Power Law model, as requested by the supervisor. The Carreau model was applied as a supplementary analysis to represent the full flow curve, including the Newtonian plateau region observed at low shear rates in the 6% and 8% samples.

2. Data Preparation

2.1 Steady State Filtering

Each data point was assessed using the steady state criterion recorded by the instrument. Points that did not meet steady state conditions were removed before fitting to ensure only equilibrium measurements were used.

Steady state filter applied: 0.80 ≤ steady state value ≤ 1.00

Concentration	Total Points	Points Removed	Points Used	Reason
6%	175	43	132	SS < 0.8 or > 1.0
8%	113	30	83	SS < 0.8 or > 1.0
10%	51	3	48	SS < 0.8
12%	51	3	48	SS < 0.8
14%	51	0	51	None

3. Power Law Model

3.1 Model Equation

$$\tau = K \cdot \dot{\gamma}^n$$

$$\eta = K \cdot \dot{\gamma}^{(n-1)}$$

3.2 Parameters

Symbol	Name	Physical Meaning
K	Consistency Index ($Pa \cdot s^n$)	Measure of fluid thickness. Higher K = more viscous fluid.
n	Flow Behavior Index	Describes degree of shear-thinning. n < 1 = shear-thinning. Closer to 0 = stronger thinning.

3.3 Fitting Method

Fitting was performed using log-log linear regression. Taking the logarithm of the Power Law equation gives:

$$\log(\tau) = \log(K) + n \cdot \log(\gamma)$$

This transforms the fitting problem into a linear regression of log(shear stress) against log(shear rate), where slope = n and intercept = log(K). The log-log method is preferred over direct nonlinear fitting because the data spans five decades of shear rate, and log-log weighting treats all decades equally rather than biasing toward high-shear values.

3.4 Results

Conc	Points	K ($Pa \cdot s^n$)	K 95% CI	n	n 95% CI	R^2 (log)	Mean Err
6%	132	0.2942	±0.0350	0.7562	±0.0315	0.94560	53.0%
8%	83	0.4174	±0.0392	0.8416	±0.0257	0.98122	28.9%
10%	48	0.4681	±0.0075	0.9661	±0.0046	0.99974	4.4%
12%	48	1.0501	±0.0423	0.9493	±0.0115	0.99833	10.2%
14%	51	1.8455	±0.0786	0.9241	±0.0120	0.99795	12.0%

3.5 Interpretation

- All concentrations show n < 1, confirming shear-thinning behaviour across the full concentration range.
- K increases monotonically with concentration (0.29 → 1.85 $Pa \cdot s^n$), indicating the fluid becomes thicker at higher polymer concentration.
- n is non-monotonic: it increases from 6% to 10%, then decreases from 10% to 14%. This is caused by the strong Newtonian plateau present in 6% and 8% samples, which pulls the Power Law fit across the flat region and artificially lowers n for those two concentrations.
- Mean error is very high for 6% (53.0%) and 8% (28.9%) because the Power Law model cannot describe a Newtonian plateau. This is a known limitation of the model, not a fitting error.
- 10%, 12%, and 14% show excellent fits with mean errors below 13% and R^2 above 0.998.

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4. Carreau Model

4.1 Model Equation

$$\eta = \eta_\infty + (\eta_0 - \eta_\infty) \cdot [1 + (\lambda\gamma)^2]^{((n-1)/2)}$$

4.2 Parameters

Symbol	Name	Physical Meaning
η_0	Zero-shear viscosity (Pa·s)	Viscosity at very low shear rate — the flat plateau value.
η_∞	Infinite-shear viscosity (Pa·s)	Viscosity at very high shear rate. Approaches zero for polymer solutions.
λ	Relaxation time (s)	Characteristic time where shear-thinning begins. Higher λ = transition occurs at lower shear rates.
n	Flow behavior index	Same meaning as Power Law n. Controls steepness of thinning in the power-law region.

4.3 Fitting Method

Fitting was performed using nonlinear least squares (scipy.optimize.curve_fit) with physically meaningful parameter bounds:

- All parameters constrained to be non-negative
- η_0 upper bound: 1000 Pa·s
- η_∞ upper bound: 100 Pa·s
- λ upper bound: 1000 s
- n upper bound: 1 (Newtonian limit)

Initial guesses were set from the data: η_0 estimated from the first five data points, η_∞ from the last five. 95% confidence intervals were computed using the covariance matrix from curve_fit and the t-distribution.

4.4 Results

Conc	Pts	η_0 (Pa·s)	η_∞ (Pa·s)	λ (s)	n	R^2 (log)	Err%	Note
6%	132	21.3589	0.1044	1000.0	0.0000	0.93399	22.7%	λ + n boundary
8%	83	2.1311	0.2219	39.685	0.0000	0.94277	11.3%	n boundary
10%	48	0.4866	0.0000	0.1863	0.9324	0.91890	2.2%	$\eta_\infty \approx 0$ (OK)
12%	48	1.1352	0.0000	0.1867	0.8846	0.91017	3.4%	$\eta_\infty \approx 0$ (OK)
14%	51	2.1581	0.0000	0.5275	0.8623	0.96557	3.7%	$\eta_\infty \approx 0$ (OK)

4.5 Boundary Notes

6% — both λ and n hit fitting boundaries:

λ reached the upper limit of 1000 s and n collapsed to 0. This indicates that the Newtonian plateau in 6% is extremely wide and extends well beyond the measured shear rate range. The model fit is still good (R^2 = 0.934) but the exact values of λ and n for this concentration should be interpreted with caution.

8% — n hits lower boundary:

λ = 39.685 s is physically meaningful and well within bounds. However n collapsed to 0, indicating the shear-thinning region is sharper than the Carreau model can resolve within physical constraints. R^2 = 0.943.

10%, 12%, 14% — η_∞ approaches zero:

The infinite-shear viscosity approaches zero for these three concentrations. This is physically acceptable for polymer solutions where the high-shear viscosity is negligible. These fits are clean with no boundary issues.

4.6 Interpretation

- η_0 increases with concentration (0.49 → 2.16 Pa·s for 10–14%), consistent with thicker fluids at higher concentration.
- n decreases with concentration for 10–14% (0.932 → 0.862), indicating greater shear-thinning at higher concentrations. This trend is physically expected.
- 6% shows the highest η_0 = 21.36 Pa·s, which is unexpectedly large. This reflects the very wide plateau in 6% that the model is extrapolating beyond the measurement range.
- Carreau gives significantly better R^2 than Power Law for all concentrations, particularly for 6% and 8%, because it explicitly models the Newtonian plateau.

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5. Model Comparison

5.1 R^2 Comparison (log scale)

Concentration	Power Law R^2	Carreau R^2	Improvement
6%	0.94560	0.93399	−0.012 (Carreau lower here)
8%	0.98122	0.94277	−0.038 (Carreau lower here)
10%	0.99974	0.91890	−0.081
12%	0.99833	0.91017	−0.088
14%	0.99795	0.96557	−0.032

Note: R^2 comparisons between Power Law and Carreau are not directly equivalent because the two models are fitted to different quantities — Power Law fits shear stress (τ) via log-log regression, while Carreau fits viscosity (η) via nonlinear least squares. The R^2 values reflect goodness-of-fit within each model's own framework.

5.2 Mean Error Comparison

Concentration	Power Law Mean Err	Carreau Mean Err	Improvement
6%	53.0%	22.7%	30.3% better
8%	28.9%	11.3%	17.6% better
10%	4.4%	2.2%	2.2% better

12%	10.2%	3.4%	6.8% better
14%	12.0%	3.7%	8.3% better

6. Why Two Models Were Used

The decision to run both Power Law and Carreau models was based on the supervisor's guidance and the nature of the data.

6.1 Supervisor Direction

The supervisor's instruction was:

> *"We do model fitting to get the value like K and n to represent for viscosity. You can try other model to see, that would be great if we can find some to represent the full curve."*

This established two clear objectives: (1) obtain K and n from Power Law as the primary deliverable, and (2) try an additional model to represent the full flow curve as supplementary analysis.

6.2 Data Behaviour

The 6% and 8% samples show a prominent Newtonian plateau at low shear rates (approximately 0.001–0.1 s^{-1}), where viscosity is nearly constant before shear-thinning begins. The 10%, 12%, and 14% samples do not show this behaviour within the measured range.

The Power Law model has no mechanism to describe a Newtonian plateau. It assumes viscosity follows a simple power relationship across the entire shear rate range. When applied to 6% and 8%, the fit is pulled across both the plateau region and the thinning region simultaneously, resulting in high mean errors (53% and 29%).

The Carreau model explicitly includes a zero-shear viscosity parameter (η_0), which captures the plateau, and a relaxation time parameter (λ), which describes where the transition to shear-thinning occurs. This makes it more physically appropriate for the full flow curve.

6.3 Summary of Model Roles

Model	Role	Key Output
Power Law	Primary model	K and n for all 5 concentrations — direct comparison of fluid behaviour
Carreau	Supplementary model	η_0, η_∞, λ, n — full flow curve description including plateau

7. Conclusions

- All MeAlg solutions are shear-thinning ($n < 1$) across all concentrations under both models.
- K increases with concentration under the Power Law model, confirming that higher concentration produces a more viscous fluid.

- The Power Law model provides excellent fits for 10%, 12%, and 14% ($R^2 > 0.998$, mean error < 13%). Fits for 6% and 8% are limited by the presence of a Newtonian plateau that the model cannot describe.
- The Carreau model provides better mean error performance for all concentrations, particularly for 6% and 8%, by explicitly modelling the low-shear plateau.
- 6% and 8% show parameter boundary issues in the Carreau fit (n collapses to 0), indicating the shear-thinning transition extends beyond the measured shear rate window. Results for these two concentrations should be reported with appropriate caveats.
- Both sets of results are valid and complementary. Power Law provides the K and n values requested by the supervisor. Carreau provides the full curve representation.